

STAT-2311
IMPORTANT TOPICS

Chapter 04:

- Correlation, correlation coefficient, properties of correlation coefficient.
- Regression, regression coefficient, properties of regression coefficient.
- Difference between correlation coefficient and regression coefficient.
- Interpretation
- Scattered diagram.

Chapter 05:

- Experiment, outcome, random experiment, sample space def.
- Classical probability, Axiomatic probability def. ***
- Additive law def.
- Bayes' Theorem & Bayes' theorem related math (slide).

Chapter 06:

- Random Variable, and example.
- Computer science-related examples of discrete and continuous random variables.
- Probability Function and Probability Density Function def + difference between them.
- Mathematical expectation def + properties.
- Variance of random variable def + properties.

Chapter 07 & 08:

- Binomial distribution
- Assumption of binomial distribution
- Poisson distribution
- Practical application of Poisson distribution (according to the CSE field).
- Normal distribution
- Importance of normal distribution.

Also,

- Importance of Correlation and regression in data analysis.
- Importance of random variable and mathematical expectation in CSE.
- Importance of Poisson distribution in CSE.
- Importance of normal distribution in CSE.

MATH: https://drive.google.com/file/d/1LaxR6FRaLKagHGo_XS3sP4fTN-jO8I-B/view?usp=sharing